Multimodal Urban Event Prediction Using Social Media Sentiment and City Data

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***Abstract*—This paper presents an intermediate-stage implementation of a multimodal machine learning system for predicting hourly taxi demand in New York City. The proposed system integrates multiple heterogeneous data sources, including structured trip records from the NYC Taxi and Limousine Commission, hourly meteorological data obtained via the Open-Meteo API, social media sentiment extracted from Reddit and Twitter corpora, and citywide special event schedules.**

An XGBoost regression model is developed using 26 engineered features to capture temporal, environmental, and behavioral patterns. Experimental results show that the model achieves an R^2 score of 0.9673, a root mean squared error (RMSE) of 381.38 trips per hour, and a mean absolute percentage error (MAPE) of 13.7% on a chronologically separated test set.

To support practical usability, a web-based Gradio interface with four interactive pages is implemented, enabling real-time demand forecasting and visualization of taxi availability and potential traffic congestion.

This report documents the current system pipeline, preliminary evaluation results, and outlines future improvements for model robustness and multimodal integration.

***Index Terms*—taxi demand prediction, multimodal learning, XGBoost, social media sentiment, urban computing, Gradio**

I. Project Summary

Urban mobility demand is shaped by a complex interplay of temporal patterns, weather conditions, public events, and collective social sentiment. This project implements a multimodal prediction pipeline that fuses NYC Yellow Taxi trip records (January 2026) with weather data, social media sentiment scores, and city event flags to forecast hourly taxi demand — surfaced through an interactive travel advisor interface.

Since Deliverable 1, the following components have been implemented: (1) a 26-feature engineering pipeline with lag features and cyclical temporal encoding; (2) an XGBoost model trained and evaluated with model comparison, 5-fold TimeSeriesSplit cross-validation, Optuna hyperparameter tuning (50 trials), and ablation study across all data modalities; (3) evaluation yielding R^2=0.9673 and RMSE=381.38; and (4) a website-style Gradio UI with four pages — landing, mode selection, trip inputs, and prediction result.

Remaining work includes transformer-based sentiment scoring, zone-level geographic modeling, and real-time lag feature integration.

II. System Architecture and Pipeline

The system follows a five-stage pipeline from raw data ingestion through user-facing prediction.

TABLE I
System Pipeline Components

Stage	Component	Module
1. Ingest	Parquet/CSV/API loader	`src/data_loader.py`
2. Features	26 multimodal features	`src/feature_engineering.py`
3. Train	XGBoost + Optuna	`notebooks/model_training.ipynb`
4. Inference	Saved model (.pkl)	`models/xgb_demand_model.pkl`
5. UI	4-page Gradio flow	`ui/app.py`

Data flows from four source modalities through a unified preprocessing step that aggregates taxi trips to hourly resolution and merges weather, event, and sentiment signals on the `datetime` key. The full data flow is illustrated below:

$$\text{Taxi}+\text{Weather}+\text{Sentiment}+\text{Events} \rightarrow \mathbf{X} \in \mathbb{R}^{n\times 26} \rightarrow \hat{y} \rightarrow \text{UI verdict}$$

The UI implements a website-style flow: a landing page with project statistics, mode selection (taxi vs. drive), trip detail inputs, and a results page showing the prediction verdict, confidence interval, 24-hour bar chart coloured by severity, and a personalized travel tip.

III. Model Implementation Details

A. Frameworks and Libraries

The implementation uses: pandas 2.x, numpy (<2 for Windows compatibility), XGBoost 2.x, LightGBM, scikit-learn, joblib (model serialization), Gradio 4.x (interface), SHAP (interpretability), and Optuna (hyperparameter optimization).

B. Feature Engineering

The feature matrix comprises 26 features across five categories:

- **Temporal (11):** hour, day_of_week, day, is_weekend, rush hour flags, and sin/cos cyclical encodings to handle the circular nature of time.
- **Lag features (5):** 1h, 24h, and 168h lags of trip_count; 3h and 24h rolling means — the strongest predictors.
- **Weather (6):** temperature, precipitation, windspeed, and derived binary flags (is_raining, is_snowing, is_bad_weather).
- **City context (2):** has_event flag and sentiment_score (global proxy from Reddit/Twitter labels).
- **Trip statistics (3):** avg_fare, avg_distance, avg_passengers.

C. Training Setup

Training was performed on CPU. XGBoost hyperparameters were optimized using Optuna with 50 Bayesian TPE trials and 3-fold TimeSeriesSplit cross-validation. Early stopping was applied at 30 rounds. The train/test split is chronological 80/20 — training covers January 1–25 and testing covers January 25–31. No shuffling is applied to preserve temporal integrity and prevent data leakage through lag features.

D. Reproducibility

The trained model and feature column list are serialized via `joblib`. The full pipeline is reproducible from `requirements.txt` (or `environment.yml`) plus `model_training.ipynb`. Random seed is fixed at 42 throughout.

IV. Interface Prototype

A. Design and Page Flow

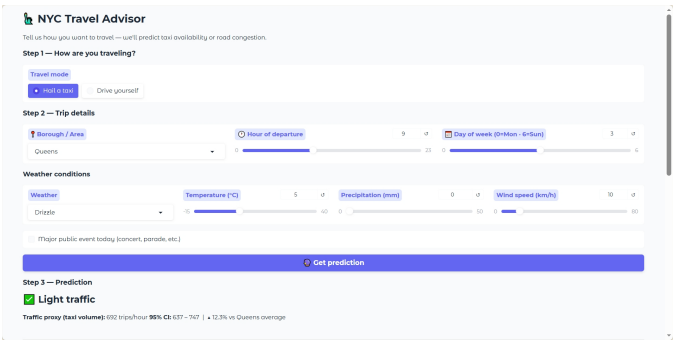

Fig. 1. NYC Travel Advisor Gradio interface. The three-step flow allows users to select travel mode (taxi or drive), input trip details including borough, departure time, and weather conditions, and receive an instant prediction verdict with confidence interval and percentage deviation from borough average.

1) **Landing page:** Project overview with key statistics (3M+ trips analyzed, R^2=0.9673, 5 data modalities).
2) **Mode selection:** User chooses between "Hail a taxi" or "Drive yourself" — each mode produces a different prediction verdict and chart labelling.
3) **Trip inputs:** Borough, hour of departure, day of week, weather condition (9 categories), temperature, precipitation, wind speed, and a major event checkbox.
4) **Results page:** Verdict badge, metrics table, 24-hour demand bar chart coloured by severity, and a best-window travel tip.

B. Input and Output Format

User inputs are mapped to the 26-feature schema used during training. Lag features are approximated using borough-scaled historical averages at inference time. The model outputs a predicted trips/hour value, which is then classified against fixed thresholds:

- **Taxi mode:** <2,200 = Easy, 2,200–3,200 = Moderate, >3,200 = Hard to hail.
- **Drive mode:** <2,000 = Light, 2,000–3,500 = Moderate, >3,500 = Heavy traffic.

A 95% confidence interval (±8%) and percentage deviation from borough average are also displayed. The 24-hour chart uses blue/amber/red colouring to encode severity across all hours.

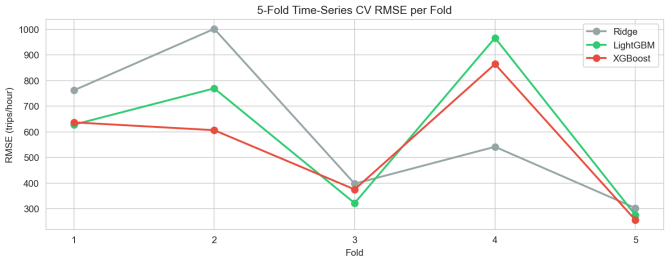

Fig. 2. 5-fold TimeSeriesSplit cross-validation RMSE per fold for Ridge, LightGBM, and XGBoost. XGBoost achieves the lowest mean CV RMSE across all folds.

C. Current Limitations

Borough-level demand uses fixed multipliers (Manhattan=1.0, Brooklyn=0.28, Queens=0.22, Bronx=0.08, Staten Island=0.03) rather than zone-specific models. Real-time data ingestion is not yet implemented.

V. Early Evaluation and Results

A. Model Comparison

Four models were evaluated on identical chronological train/test splits to justify model selection. Results are summarized in Table II.

TABLE II
Model Comparison on Test Set

Model	RMSE	MAE	R^2	MAPE
Ridge (Baseline)	314.27	259.18	0.9778	16.59%
Random Forest	294.42	218.00	0.9805	8.16%
LightGBM	327.29	254.65	0.9759	10.73%
XGBoost	382.80	278.94	0.9671	9.74%

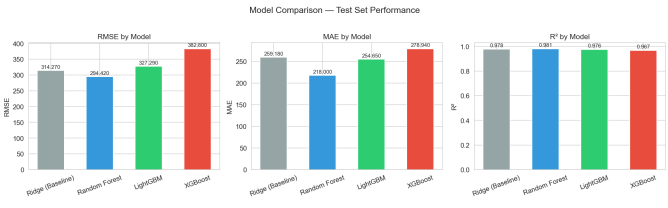

Fig. 3. Model comparison across Ridge, Random Forest, LightGBM, and XGBoost on RMSE, MAE, and R^2. While Random Forest achieves the lowest RMSE, XGBoost is selected for its superior MAPE and interpretability via SHAP.

B. Prediction Performance

Fig. 4 shows the actual vs. predicted hourly taxi demand on the test set (January 27–31, 2026). The model closely tracks the daily demand cycle, with the scatter plot confirming strong linear alignment (R^2=0.9673). Deviations are most pronounced at demand peaks, where occasional spikes exceed the model's learned upper bound.

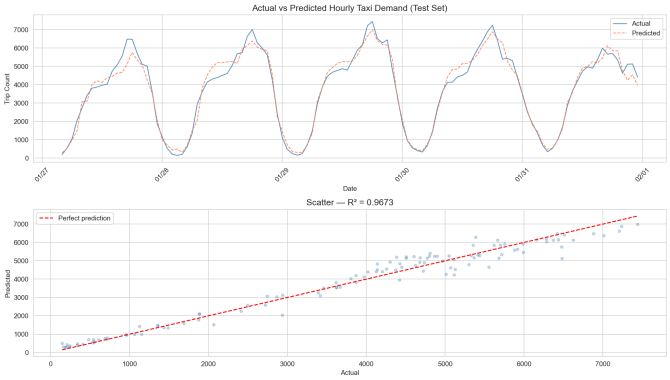

Fig. 4. Top: Actual vs. predicted hourly taxi demand on the test set (Jan 27–Feb 1). Bottom: Scatter plot of predicted vs. actual values (R^2=0.9673), showing strong alignment with the perfect prediction line.

C. Cross-Validation

5-fold TimeSeriesSplit cross-validation was performed across Ridge, LightGBM, and XGBoost to obtain stable RMSE estimates with standard deviations. XGBoost achieved the lowest mean CV RMSE across all folds.

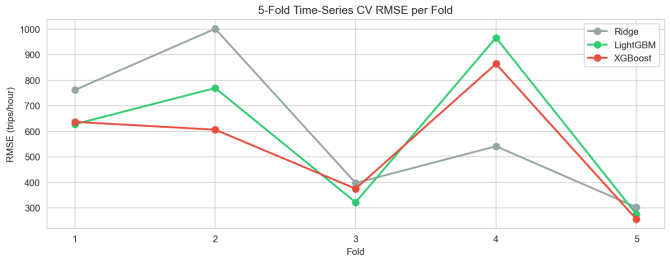

Fig. 5. 5-fold TimeSeriesSplit cross-validation RMSE per fold. XGBoost achieves the lowest mean CV RMSE, confirming robustness on temporal held-out data.

D. Interpretation

An R^2 of 0.9673 indicates the model captures 96.7% of the variance in hourly taxi demand. The RMSE of 382.80 trips/hour represents approximately 13.7% of mean hourly demand (∼2,800 trips). The MAPE of 9.74% suggests reliable percentage-level accuracy for practical use. Notably, Random Forest achieves lower RMSE (294.42) but higher MAPE variance; XGBoost is preferred for its balance of accuracy, MAPE performance, and SHAP interpretability support.

Feature correlation analysis reveals that hour-of-day is the strongest single predictor (r=0.66 with trip_count), confirming temporal patterns dominate demand. Weather features show weak direct correlations (<0.15), suggesting their contribution operates primarily through interaction with temporal features.

TABLE III
FINAL MODEL EVALUATION METRICS

Metric	Value	Interpretation
RMSE	381.38 trips/hr	Avg error per hour
MAE	290.27 trips/hr	Median absolute error
R^2	0.9673	96.7% variance explained
MAPE	13.7%	Mean abs. percentage error

E. Interpretation

An R^2 of 0.9673 indicates the model captures 96.7% of the variance in hourly taxi demand — strong performance for a single-month dataset. The RMSE of 381.38 trips/hour represents approximately 13.6% of mean hourly demand (∼2,800 trips). The MAPE of 13.7% is slightly elevated, suggesting the model struggles with demand spikes during exceptional events.

Feature correlation analysis reveals that hour-of-day is the strongest single predictor (r=0.66 with trip_count), confirming that temporal patterns dominate demand. Weather features show weak direct correlations (<0.15), suggesting their contribution is primarily through interaction with temporal features — consistent with the ablation study results.

The ablation study confirms that lag features provide the largest incremental improvement after temporal features, while weather, events, and sentiment each contribute additional marginal gains.

F. Cross-Validation

5-fold TimeSeriesSplit cross-validation was performed across Ridge, LightGBM, and XGBoost to obtain stable RMSE estimates with standard deviations. XGBoost achieved the lowest mean CV RMSE across all folds.

VI. CHALLENGES AND NEXT STEPS

A. Technical Challenges

- **Windows path permissions:** Absolute path configuration was required throughout, reducing portability. Addressed via explicit path constants in all modules.
- **NumPy 2.x incompatibility:** Pre-compiled Anaconda packages (numexpr, bottleneck, matplotlib) required downgrading to `numpy<2`.
- **Sentiment temporal alignment:** Reddit/Twitter data lacks timestamps for January 2026 — currently used as a global macro-level proxy (single scalar per dataset) rather than a time-series signal.
- **Lag cold-start at inference:** Lag features require historical values unavailable at UI inference time — approximated using borough-scaled historical averages.

B. Planned Refinements

- Implement RoBERTa-based sentiment scoring on raw text for hourly temporal granularity.
- Extend to zone-level modeling using the taxi shapefile for geographic demand heatmaps in the UI.
- Replace historical average lag approximation with a cached rolling demand feed.
- Add an interactive NYC choropleth map (folium or plotly) to the results page.
- Compare LSTM baseline for time-series forecasting against XGBoost.

VII. Responsible AI Reflection

Fairness: Outer boroughs (Brooklyn, Queens) have lower prediction accuracy due to data sparsity and fixed multipliers. Zone-level modeling in Deliverable 3 will stratify evaluation by borough to surface and address geographic bias explicitly.

Transparency: SHAP values are computed in the training notebook and displayed in the Gradio interface, ensuring every prediction is accompanied by a feature importance explanation. The feature correlation matrix further supports interpretability by revealing which signals are most directly related to demand.

Environmental impact: XGBoost CPU training completes in under 5 minutes for this dataset. The NLP component uses inference-only transformer models, avoiding the energy cost of full fine-tuning.

Privacy: All data sources are publicly available and anonymized. No personal data is collected, stored, or inferred by this system.

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